Wall256

SU(3) Yang-Mills Strong-Coupling Conditional Reduction

A Complete Technical Report

History -- Architecture -- Barrier -- Experiments -- Conjectures -- Audit

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Opera Numerorum | June 2026 | Lean 4 / Mathlib v4.12.0

HONESTY NOTICE -- READ FIRST This report makes NO claim that the Yang-Mills mass gap is proven. YM_STATUS = OPEN. Clay Surface #1: OPEN. No mu > 0. No spectral gap. SORRY = 0. AXIOMS = {propext, Classical.choice, Quot.sound}. The entire mathematical content rests on THREE named open hypotheses (H1, H2, H3). None is discharged. The conclusion holds ONLY IF all three hold. Commit: 8eeab54 | Towers/YM/Wall256_Scaffold.lean | NOT a brick

AUDIT FINDINGS (caught during pre-build certification -- June 2026) AUDIT-1: P5_genuine = 1000000001119 = 7 x 142857143017 -- COMPOSITE, NOT PRIME. TOWERS_YM_v2.3 claimed this was the '13-digit boundary prime'. It is not prime. Remarkable: 142857143017 begins with 142857, the cyclic decimal of 1/7 (the KP threshold). The boundary number is divisible by the threshold denominator 7. AUDIT-2: beta_0 bracket: our 7-moment partial sum cannot reproduce the CERT_Arb bracket; full 36-moment computation (performed in source workspace) is authoritative. CERT_Arb SHA: b5a9f0a7666a91f283a7d4531ae99dff2097c2cedef10424b77833b7bbc840d3

`COMMIT = 8eeab54 | SORRY = 0 | YM_STATUS = OPEN`

Contents

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1. The Historical Problem: Yang-Mills Mass Gap

The Yang-Mills existence and mass gap problem -- one of the seven Clay Millennium Prize Problems -- asks, for a compact simple Lie gauge group G (here G = SU(3), the gauge group of QCD), for the construction of a quantum Yang-Mills theory on R^4 satisfying the Wightman axioms, whose Hamiltonian has a STRICTLY POSITIVE spectral gap above the vacuum.

The mass gap $m > 0$ explains why the strong nuclear force is short-range despite being classically mediated by massless gluons -- the quantum theory acquires a mass through confinement. Proving this rigorously from first principles is the problem.

Three Equivalent Formulations

```
(E1)  Two-point exponential clustering:
      |<W(C1) W(C2)> - <W(C1)><W(C2)>| <= C * exp(-Delta * dist(C1, C2)),  Delta > 0
      W(C) = Wilson loop (gauge-invariant trace of holonomy around C)
      <.> = expectation in the Euclidean lattice YM measure

(E2)  Transfer-operator spectral gap:
      spec(T_real | H_0) in [0, lambda_1],   lambda_1 < 1
      Delta = -log(lambda_1) > 0
      T_real = Wilson/heat-kernel transfer operator (one Euclidean time step)

(E3)  HasMassGap predicate:
      <x, T x> >= (1-m)||x||^2  for all x in H_0,   m > 0
      Equivalently: lambda_1 <= exp(-m)
```

The HONEST GAP: a non-vacuous discharge of (E1)-(E2)-(E3) requires a REAL SU(3) Wilson transfer operator with a verified T_real > 0. The ingredients -- SU(3) character theory and a verified heat-kernel transfer operator with explicit spectral bounds -- are absent from mathlib v4.12.0. Every YM module in this repository that claimed to touch (E1)-(E3) was discharged at a degenerate witness (see Section 11).

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2. The Architecture: Everything Reduces to One Inequality

The Osterwalder-Seiler strong-coupling analysis provides a rigorous lattice route to exponential clustering. The entire chain of implications collapses to a single condition on the single-site Haar weight:

THE HINGE INEQUALITY: w1 < 1/7 w1 = integral_{SU(3)} exp(-beta * S) d(Haar), S = (3 - Re tr U) / 3 If w1 < 1/7, the polymer entropy series converges (KP criterion), and the lattice two-point function decays exponentially with rate Delta > 0. This requires beta > beta_0 = 2.079416880123... (CERT_Arb certified).

The Reduction Tower

```
w1 < 1/7                                                    [H1] OPEN
single-site Haar weight is strictly small

=> a(n) <= exp(-I * n),  I > log(7) = 1.9459...            [H2] OPEN
Osterwalder-Seiler 1978 Thm 2.1: single-site smallness
propagates to per-polymer activity rate I > log(7)

=> sum_n N(n)*a(n) <= sum_n 7^n * exp(-I*n)                PROVED
                     = sum_n (7*exp(-I))^n  <  infinity
Comparison test converges since I>log(7) <=> 7*exp(-I)<1
[kp_summable_of_truncatedActivity -- machine-checked]

=> 0 < rho < 1, |corr(x,y)| <= C * rho^{sep(x,y)}         [H3] OPEN
Brydges-Federbush: KP summability => geometric two-point decay

=> Delta = -log(rho) > 0, |corr| <= C * exp(-Delta*sep)     PROVED
rho^d = exp(-Delta*d) algebra
[mass_gap_pos_of_spectral_gap -- machine-checked]

CONDITIONAL CONCLUSION: abstract lattice two-point decay
Holds ONLY IF H1 + H2 + H3 are ALL discharged.
This is NOT a mass gap, NOT a Clay result, NOT a spectral gap.
```

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3. The Wall256 Theorem -- Verbatim Lean (commit 8eeab54)

```
-- Towers/YM/Wall256_Scaffold.lean  (commit 8eeab54)
-- SORRY: 0  |  AXIOMS: classical trio  |  NOT a brick  |  YM_STATUS: OPEN

theorem strong_coupling_decay_of_open_inputs
     {E : Type*} (corr sep : E -> E -> R) (C rho w1 : R)
     {N a : N -> R}
     (hN0  : forall n, 0 <= N n)
     (hN   : forall n, N n <= (7 : R) ^ n)
     (hw1  : w1 < 1 / 7)                                -- H1  OPEN
     (hOS  : w1 < 1 / 7 -> TruncatedActivityBound a)    -- H2  OPEN
     (h_bridge : Summable (fun n : N => N n * a n) ->   -- H3  OPEN
           0 < rho /\ rho < 1 /\
           forall x y, |corr x y| <= C * rho ^ (sep x y)) :
     exists Delta : R,
       0 < Delta /\ forall x y,
         |corr x y| <= C * Real.exp (-Delta * sep x y) :=
   su2_gap_of_truncatedActivity corr sep C rho hN0 hN (hOS hw1) h_bridge
```

Symbol	Meaning
E : Type*	Abstract type -- ANY space of sites. No real lattice built.
corr, sep	Abstract correlator and separation. No real Wilson loop.
N n <= 7^n	Polymer entropy: connected polymers of size n number at most 7^n (SU(3) coordination).
hw1 : w1 < 1/7	H1 OPEN -- single-site Haar weight strictly below the KP threshold.
hOS	H2 OPEN -- Osterwalder-Seiler: single-site smallness => activity rate I > log(7).
h_bridge	H3 OPEN -- Brydges-Federbush: KP summability => geometric two-point clustering.
TruncatedActivityBound a	(forall n, 0<=a n) AND (exists I, log(7)<I AND forall n, a n<=exp(-I)^n)
su2_gap_of_truncatedActivity	Already-proved combinator in Wall256_Note.lean. Group-agnostic. Asserts no gap.
Conclusion	EXISTS Delta>0 s.t. |corr|<=C*exp(-Delta*sep). CONDITIONAL on H1+H2+H3 only.

SORRY = 0. Proof body is one line: plug H1, H2, H3 into the proved combinator.

No real SU(3) operator, no real Wilson loop, no real lattice metric is built.

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4. The Three Open Inputs: H1, H2, H3

H1 -- SU(3) single-site Haar weight (strict bound)

```
hw1 : w1 < 1/7
w1 = integral_{SU(3)} exp(-beta * S) dHaar,  S = (3 - Re tr U)/3
```

The single-site Boltzmann weight w1(beta) decreases as beta increases (Laplace transform of a non-degenerate non-negative random variable -- strictly monotone). The strict bound w1 < 1/7 holds for beta > beta_0, certified at beta_0 in [2.079416880123, 2.079416880124] (CERT_Arb). Mathlib v4.12.0 cannot evaluate SU(3) Haar integrals, so this stays a named hypothesis on abstract w1.

STATUS: OPEN. Source: SU(3) character theory / verified cubature. Absent from mathlib v4.12.0.

H2 -- Osterwalder-Seiler 1978, Theorem 2.1 (Ursell / cluster step)

```
hOS : w1 < 1/7 -> TruncatedActivityBound a
```

Single-site smallness propagates, via the truncated (Ursell) cluster expansion, to a per-size connected-polymer activity bound: a(n) <= exp(-I*n) with I > log(7) STRICTLY. The cluster expansion is absent from mathlib. The strict rate I > log(7) (not >=) is essential -- see Section 5.

STATUS: OPEN. Source: Osterwalder & Seiler, Ann. Phys. 110 (1978), Thm 2.1. Absent from mathlib v4.12.0. Hardest open leaf.

H3 -- Brydges-Federbush KP Bridge

```
h_bridge : Summable (N n * a n) ->
           0<rho AND rho<1 AND forall x y, |corr x y| <= C * rho^(sep x y)
```

Kotecky-Preiss summability of the entropy-weighted polymer series implies geometric two-point clustering with spectral radius rho < 1. Standard cluster-expansion theory (Friedli & Velenik, Stat. Mech. of Lattice Systems, Ch. 5), absent from mathlib v4.12.0. Carried as a hypothesis, NOT by sorry.

STATUS: OPEN. Source: Friedli & Velenik 2018, Ch. 5. Absent from mathlib v4.12.0.

---

5. Why the Bound Must Be Strict: w1 < 1/7 (not <=)

The strictness is not a formality -- it is the entire reason the polymer entropy series converges. The boundary case w1 = 1/7 makes the comparison test fail (geometric series ratio = 1, series diverges).

Case	Rate	Geometric ratio	Series	Verdict
w1 < 1/7	I > log(7)	7*exp(-I) < 1	sum (7*exp(-I))^n	CONVERGES (KP ok)
w1 = 1/7 (boundary, beta=0.85 excluded)	I = log(7)	7*exp(-I) = 1	sum 1^n = sum 1	DIVERGES (vacuous)
w1 > 1/7	I < log(7)	7*exp(-I) > 1	grows without bound	DIVERGES

CERTIFIED: Convergence vs divergence at the boundary

```
log(7)       = 1.945910149055313
exp(-log(7)) = 0.142857142857143 = 1/7  [VERIFIED: identical to 15dp]

BOUNDARY (I = log 7): ratio = 7 * exp(-log7) = 1.000000000000000  => DIVERGES

I = log(7) + 0.001:  ratio = 0.999000  sum = 1000.5   [CONVERGES]
I = log(7) + 0.010:  ratio = 0.990050  sum =  100.5   [CONVERGES]
I = log(7) + 0.100:  ratio = 0.904837  sum =   10.5   [CONVERGES]
I = log(7) + 0.500:  ratio = 0.606531  sum =    2.5   [CONVERGES]

Heuristic at beta=2.07942 (just above beta_0):
w1 ~ 0.142853 < 1/7 = 0.142857  =>  I = -log(w1) ~ 1.945960 > log(7) = 1.945910
I - log(7) ~ 0.000050  (very narrow strict margin just above threshold)
```

Strict inequality I > log(7) required and verified at beta > beta_0.

Boundary beta = 0.85 (w1 = 1/7) is EXCLUDED -- the reduction is vacuous there.

---

6. The beta_0 Certificate -- SU(3) Haar Integral

STATUS: VERIFIED_OUT_OF_TOWER (CERT_Arb, authoritative) NOT Lean. NOT trio-clean. Numerics only. Discharges ZERO Lean obligations. AUDIT-2 NOTE: Our 7-moment partial sum (m_0..m_6) cannot reproduce the CERT_Arb bracket. The full computation uses N=36 exact moments via constant-term extraction over the SU(3) torus. That computation was performed in the source workspace (TheoremaAureum143). We certify the qualitative results below; CERT_Arb is the authoritative enclosure.

The SU(3) Haar Integral Formula

```
w1(beta) = exp(-beta) * sum_{n=0}^{inf} (beta/3)^n * m_n / n!

m_n = <(Re tr U)^n>_{Haar on SU(3)}  -- exact rational moments
computed by constant-term extraction over the SU(3) torus
(Weyl integration formula)

Rigorous tail bound:  |R_N| <= beta^{N+1} / (N+1)! / (1 - beta/(N+2))
(from |Re tr U| <= 3 => |m_n| <= 3^n)

CERT_Arb: N=36 terms, tail <= 4.46e-32  (mpmath.iv dps=80, outward rounding)
```

Exact Moments m_n (from CERT_Arb -- machine-certified rationals)

n	m_n (exact rational)	m_n (decimal)
0	1	1.000000000000000
1	0	0.000000000000000 (SU(3) symmetry)
2	1/2	0.500000000000000
3	1/4	0.250000000000000
4	3/4	0.750000000000000

5	15/16	0.937500000000000
6	65/32	2.031250000000000

Certified w1 sign table (CERT_Arb, N=36, dps=80)

beta	w1(beta) enclosure	vs 1/7	Verdict
0.86	[0.432367..., 0.432367...]	> 1/7 = 0.142857...	D4 FAILS -- w1 >> 1/7
2.07941	[0.142857993..., 0.142857994...]	> 1/7	Above threshold
2.07942	[0.142856757..., 0.142856758...]	< 1/7	Below threshold
2.079416880123	certified > 1/7	> 1/7	Low bracket endpoint
2.079416880124	certified < 1/7	< 1/7	High bracket endpoint
beta_0 in [2.079416880123, 2.079416880124]	--	CROSSING CERTIFIED	LOCKED

D4 at beta=0.86: w1(0.86) > 1/7 CERTIFIED. D4 FAILS.

beta_0 in [2.079416880123, 2.079416880124]: CERTIFIED (CERT_Arb).

Monotonicity: w1(beta) = E[exp(-beta*S)] is strictly decreasing (Laplace transform of non-degenerate S >= 0). Crossing is unique.

This does NOT produce a Lean proof object. hw1 remains an open named hypothesis in Wall256_Scaffold.lean.

7. The Bessel I Barrier -- Where mathlib Runs Out

The Haar integral w1(beta) -- the hinge of the entire Wall256 reduction -- is computed via the SU(3) character expansion, which involves modified Bessel functions of the first kind. This is the point where the formal proof hits a genuine barrier: mathlib v4.12.0 has no SU(3) Haar measure, no character theory, and no Bessel function library.

How Bessel Functions Enter

```
SU(3) heat kernel at the identity:
K_beta(I) = sum_{(m,n) in N^2} dim(m,n)^2 * exp(-beta * C2(m,n))

dim(m,n) = (m+1)(n+1)(m+n+2)/2      (Weyl dimension formula)
C2(m,n)  = m^2 + n^2 + mn + 3m + 3n (quadratic Casimir, un-normalized)

Character expansion of Re tr U on SU(2) analogy:
<exp(i*theta)>_{SU(2)} involves I_0(beta) and I_1(beta)
where I_k(x) = sum_{j=0}^{inf} (x/2)^{2j+k} / (j! * (j+k)!)

For SU(3): moments m_n = <(Re tr U)^n> computed by constant-term
extraction over the maximal torus via the Weyl integration formula.
The closed-form expressions involve SU(3) Bessel/character integrals.
```

Certified Bessel I values (verified against Z_BESSEL_I_COMPLETE CSV)

n \ x	x = 0.5	x = 1.0	x = 2.0	x = 5.0	x = 10.0
I_0	1.063483370741	1.266065877752	2.279585302336	27.239871823604	2815.716628466
I_1	0.257894305391	0.565159103992	1.590636854637	24.335642142451	2670.988303701
I_2	0.031906149178	0.135747669767	0.688948447699	17.505614966624	2281.518967726
I_3	0.002645111969	0.022168424924	0.212739959240	10.331150169151	1758.380716611

ALL 20 Bessel I values verified against Z_BESSEL_I_COMPLETE CSV: all PASS (relative error < 1e-10). mpmath computation is deterministic.

What is absent from mathlib v4.12.0

Missing component	Status	Impact
SU(3) Haar measure	ABSENT	Cannot state hw1 as a Lean term
SU(3) character theory	ABSENT	Cannot compute m_n in Lean
Modified Bessel functions I_k	ABSENT	No Bessel-based bound possible
Weyl integration formula	ABSENT	No torus decomposition of Haar
w1(beta) < 1/7 proof	ABSENT	H1 / hw1 stays open hypothesis

The Bessel I barrier is the primary reason H1 (hw1) cannot be discharged in Lean with the current mathlib. All numerical evidence is consistent with the claimed threshold; the formal proof gap is analytic, not arithmetic.

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8. Surface #1 Installment A -- Peter-Weyl Infrastructure

While the full Haar integral remains outside mathlib, the Peter-Weyl heat-kernel series HAS been formalized. Two lemmas provide the summability infrastructure needed by any cluster-expansion attack on the mass gap.

Formalized Results (8 bricks, tasks #154 and #155)

Theorem	Lean name	Statement	Status
3.1	PeterWeyl_Summable_SU3	For every t > 0: dim(m,n)^2 * exp(-t*C2(m,n)) is Summable over (m,n) in N^2.	PROVED
3.2	Weyl_sum_SU3_le_envelope	For every t > 0, N in N: S_N(t) <= K(t) (truncation <= full sum).	PROVED
3.3	Heat_kernel_envelope_ge_one	For every t > 0: K(t) >= 1 (trivial rep (0,0) contributes 1).	PROVED

Verified S_N(t) convergence (computed, not proved -- confirms formulas)

```
S_N(t=1), N= 5: sum = 1.340902  (21 pairs)
S_N(t=1), N=10: sum = 1.340902  (66 pairs)   -- converged at N=5
S_N(t=1), N=20: sum = 1.340902  (231 pairs)  -- stable to 6 dp
S_N(t=1), N=50: sum = 1.340902  (1326 pairs) -- no further contribution

dim(0,0)=1, C2(0,0)=0: PASS  dim(1,0)=3, C2(1,0)=4: PASS
dim(0,1)=3 (anti-fundamental): PASS
```

Open gaps

Task	Description	Blocks
#156 OPEN	Varadhan small-t asymptotic: K(t) <= C*t^{-4}*exp(-c/t)	Per-plaquette activity bound, all of Surface #2
#157 OPEN	Tighter Casimir: C2(m,n) >= c*(m+n)^2 as Lean lemma	Tech-debt; shortens Surface #1 proofs only

460 trio-clean bricks total. 8 bricks in PeterWeyl.lean + PeterWeylHeat.lean.

Genesis seal: eecbcd9a540aa7a2c90edd23827c73e4d1bb5af641d352f70a5de849b21f875f

9. Surface #2 -- Polymer Expansion Scaffold

Surface #2 describes the polymer-expansion route to a convergent cluster expansion for 4D SU(3) lattice YM at small coupling. It is the intended consumer of the Peter-Weyl infrastructure from Surface #1. Every central inequality is OPEN.

```
STEP 1: Per-plaquette exponential activity  [OPEN -- Lean sorry]
exists_c_per_plaquette_pw:
plaquette_activity_pw(beta,N,p) <= C * beta^{-4} * exp(-c/beta)
[blocked by Varadhan asymptotic, Task #156]

STEP 2: Kotecky-Preiss criterion  [OPEN -- ~60 sorries in ClusterExpansion.lean]
kotecky_preiss_criterion:
exists mu s.t. sum_{gamma':gamma'capta} w(gamma')*exp(mu(gamma')) <= mu(gamma)

STEP 3: Cluster expansion of log Z_Lambda  [OPEN -- not formalized]
log Z_Lambda(beta) = sum_X Phi_T(X) * prod_{gamma in X} w(gamma)

STEP 4: Area-law Wilson loop decay  [OPEN -- Wilson loops not defined]
|<W_C>| <= K * exp(-m * Area(C))

CONTINUUM LIMIT  a -> 0  [SEPARATE SURFACE -- not attempted]
```

YM tower Status: OPEN. No step above is formalized. ~60 sorries remain in Towers/Attempts/ClusterExpansion.lean.

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10. The Z-Experiments -- Numerical Testing

The Z-experiment harness tests two computation modes: T=1 (tool-assisted, deterministic) and T=0 (LLM-direct prediction). The harness measures reliability, not mathematical content. All T=0 Bessel runs are NOT_RUN_NO_API in this batch.

Z_POLYMER: LLM-direct vs tool-assisted on bit-string polymers

Input	L	Zero-run	T=1 errors	T=0 errors
10111111	8	1	0/100 (0.00)	78/100 (0.78)
10000111	8	4	0/100 (0.00)	100/100 (1.00)
1001111111111111	16	2	0/100 (0.00)	52/100 (0.52)
1000000001111111	16	8	0/100 (0.00)	100/100 (1.00)
100011...11 (24b)	24	3	0/100 (0.00)	100/100 (1.00)
100000...11 (24b)	24	12	0/100 (0.00)	100/100 (1.00)
1000001...11 (40b)	40	5	0/100 (0.00)	100/100 (1.00)
1000000...11 (40b)	40	20	0/100 (0.00)	100/100 (1.00)

Z_BOSTCONNES: LLM-direct vs tool-assisted on KMS state

beta_KMS	T=1 errors/100	T=0 errors/100
1.5	0 (0.00)	100 (1.00)
2.0	0 (0.00)	76 (0.76)
2.5	0 (0.00)	24 (0.24)
3.0	0 (0.00)	100 (1.00)
5.0	0 (0.00)	73 (0.73)
8.0	0 (0.00)	30 (0.30)

Experiment	T=1 result	T=0 result	Finding
Z_BESSEL_I (20 cells)	0/400 errors	NOT_RUN (NO-API)	Bessel I: mpmath is exact and deterministic
Z_POLYMER (8 inputs)	0/800 errors	52-100% errors	LLM unreliable for long zero-run bit strings
Z_BOSTCONNES (8 cells)	0/800 errors	24-100% errors	LLM erratic for KMS evaluation (no pattern)

Z_POLYMER_RESULTS (101)	0 errors	NOT_RUN	Tool-assisted reproduction fully deterministic

T=1 (tool-assisted): 0 errors across ALL experiments. Deterministic.

T=0 (LLM-direct): 24-100% error rates. Error rate increases with zero-run length and bit count. Not suitable for formal certification.

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11. The 13-Module Witness-Collapse Audit (PDF 263, 2026-06-03)

On 2026-06-03 a repair audit converted all 13 YM/OS modules from 'proved theorems' to named open Props and de-registered them from BRICKS. Reason: every module was discharged at a DEGENERATE WITNESS (witness collapse), not at any real SU(3) operator with T_real > 0.

The three collapse shapes

SHAPE 1 -- Trivial satisfiability: 'EXISTS C>0 AND mu>0' witnessed by C=mu=1. Says nothing about any correlator.

SHAPE 2 -- Reflexive bound: integrated_tail(L,m) := exp(-m*L), so 'bound' exp(-m*L) <= exp(-m*L) closes by le_refl. No real heat-trace integral.

SHAPE 3 -- Unused hypothesis: mass gap hypothesis assumed, then f(t)=exp(-m*t) discharged directly. Spectral hypothesis never used.

#	File	Named Open Prop	Collapse	T_real>0
1	YM/ClusteringCore	clusters_zero_OPEN	const-zero pair	none
2	YM/MassGapStandin	massGap_standin_OPEN	zero CLM on C	none
3	YM/SpectralGapCore	hasMassGap_zero_OPEN	T:=0 (vacuous)	none
4	YM/TransferOperatorBound	transfer_gap_zero_OPEN	degen. operators	none
5	YM/TwoPointDecay	clustering_zero_OPEN	const-zero fn	none
6	YM/MassGapFromDecay	mass_gap_from_zero_OPEN	zero witness op	none
7	YM/IntegratedTailReal	integrated_tail_le_exp_OPEN	reflexive bound	none
8	YM/TransferGapReal	transfer_gap_real_OPEN	degen. witness	none
9	YM/MassGapReal	mass_gap_from_transfer_OPEN	zero on H:=C	none
10	YM/ClusteringImpliesGap	clustering_implies_gap_OPEN	zero antecedent	none
11	YM/TransferImpliesClustering	transfer_implies_OPEN	const-zero fn	none
12	YM/TailImpliesTransfer	tail_implies_transfer_OPEN	definitional rewrap	none
13	YM/GapToDecay	gap_to_decay_OPEN	hypoth. unused	none

Property	Value
Modules audited	13
Non-vacuous T_real > 0 proofs	0
Converted to named open Prop	13
sorry / sorryAx / admit count	0 / 0 / 0
Axiom footprint (each)	{propext, Classical.choice, Quot.sound}
Compile (direct-lean, all modules)	EXIT 0
Surface #1 (Clay YM mass gap)	OPEN

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12. The Conjecture Landscape -- Open Problems

The Wall256 project surfaces a rich landscape of genuine open conjectures. This section catalogues them with dependencies and what would be needed to resolve them -- from analytic (Varadhan) to number-theoretic (C13 law).

Conjecture C1: hw1 Bridge

STATEMENT: For all beta > 2.079416880124: w1(beta) < 1/7 AS A LEAN PROOF TERM.

WHAT IS NEEDED: Requires SU(3) character theory or verified cubature in mathlib v4.12.0. Numerically certified (CERT_Arb). Currently absent from mathlib.

IF CLOSED: Would discharge H1. Still needs H2 (hOS) and H3 (h_bridge).

Conjecture C2: Osterwalder-Seiler Rate

STATEMENT: TruncatedActivityBound(a) with I > log(7) STRICTLY, given w1 < 1/7.

WHAT IS NEEDED: Requires formalizing Osterwalder-Seiler 1978 Thm 2.1 in Lean. The Ursell/truncated expansion is absent from mathlib. Hardest open leaf.

IF CLOSED: Would discharge H2. Still needs H3 and stays at lattice scope only.

Conjecture C3: Varadhan Small-t Asymptotic (Task #156)

*STATEMENT: K(t) <= C * t^{-4} * exp(-c/t) for t in (0, t_0] with C, c, t_0 > 0 explicit.*

WHAT IS NEEDED: Classical (Molchanov/Varadhan for compact semisimple Lie groups, exponent -dim(G)/2 = -4 for SU(3)). No machine-checked proof exists in this project.

IF CLOSED: Closes gap between Surface #1 (done) and Surface #2 (blocked on this).

Conjecture C4: Per-Plaquette Activity Bound

*STATEMENT: EXISTS C, c, t_0 s.t. plaquette_activity_pw(beta,N,p) <= C*beta^{-4}*exp(-c/beta).*

WHAT IS NEEDED: Direct consumer of Varadhan (C3). Lean: exists_c_per_plaquette_pw (sorry).

IF CLOSED: Required for Kotecky-Preiss (C5).

Conjecture C5: Kotecky-Preiss Criterion at Small Coupling

STATEMENT: EXISTS beta_0 s.t. for beta >= beta_0: KP condition holds with mu(gamma) = a|gamma|.*

WHAT IS NEEDED: Lean: kotecky_preiss_criterion (sorry, ClusterExpansion.lean line ~601). Needs polymer graph branching count and partial-sum estimate (neither formalized).

IF CLOSED: Would give convergence of the polymer log-partition series.

Conjecture C6: Area-Law Wilson Loop Decay

*STATEMENT: || <= K * exp(-m * Area(C)) for contractible Wilson loops.*

WHAT IS NEEDED: Wilson loops are not defined anywhere in the codebase. Standard result from convergent cluster expansion at strong coupling (lattice only).

IF CLOSED: Lattice area law only. Continuum limit a->0 is a SEPARATE surface.

Conjecture C7: H4 / C13 Law (Requires Correction -- see Section 13)

STATEMENT: digit_len(p) >= 13 implies Sym(p) = 1 FOR PRIME p.

WHAT IS NEEDED: Empirical: 6/6 cases. Named open Prop C13_Law_Open in H4_Derivation.lean. AUDIT-1 (Section 13): P5_genuine = 1000000001119 = 7 x 142857143017 is COMPOSITE. The law needs reformulation over primes only, and P5 needs a replacement.

IF CLOSED: After correction: law statement stays, but P5_genuine must be replaced.

Conjecture C8: Bessel I Formal Bound

*STATEMENT: I_n(x) <= exp(x) / sqrt(2*pi*x) for large x (standard asymptotic).*

WHAT IS NEEDED: No Lean formalization in this project. Would provide a formal route to bounding the SU(3) character integrals and eventually to C1 (hw1 bridge).

IF CLOSED: Does not by itself prove w1 < 1/7 as a Lean term; feeds toward C1.

Conjecture C9: Tighter Casimir Bounds (Task #157)

STATEMENT: C2(m,n) >= c(m+n)^2 and dim(m,n) = O((m+n)^3) as Lean lemmas.*

WHAT IS NEEDED: Current formalization uses only linear/product-square slack versions. Tighter bounds are standard in the informal literature.

IF CLOSED: Tech-debt only. Would shorten Surface #1 proofs; no new theorem content.

---

13. Audit Findings (caught during pre-build certification)

The Opera Numerorum rule: errors are certified and documented, never silently corrected. The following two findings were discovered by running the pre-build certification script (certificates/certify_wall256_ym.py) against the uploaded source documents.

AUDIT-1: P5_genuine is COMPOSITE (not prime)

FINDING: P5_genuine = 1000000001119 = 7 x 142857143017. NOT PRIME.

TOWERS_YM_v2.3 states: 'P5_genuine = 1000000001119 is the ONLY live P5 in v2.3 -- the real 13-digit boundary prime (digit_len = 13, Sym = 1).'

CERTIFICATION RESULT (Miller-Rabin, witnesses [2,3,5,7,11,13,17,19,23,29,31,37]):

```
P5_genuine = 1000000001119
P5_genuine % 7 = 0
P5_genuine = 7 x 142857143017
Verification: 7 x 142857143017 = 1000000001119  [CONFIRMED]
Is 142857143017 prime? YES (Miller-Rabin)
CONCLUSION: P5_genuine = 7 x (prime)  -->  COMPOSITE.
```

The digit-count claim is still correct (13 digits, PASS). The primality claim is FALSE.

The Remarkable Structure of the Factorization

The factorization P5_genuine = 7 x 142857143017 has a striking feature:

```
1/7 = 0.142857142857142857...  (the KP threshold denominator, repeating)

142857143017 begins with:  142857...
The cyclic group of 1/7 = {142857, 285714, 428571, 571428, 714285, 857142}

q = 142857143017 ~ 142857 x 10^6 + 143017
   = 142857 x (1000000 + 1.001...)  -- very close to 142857 * 10^6

KP threshold: w1 < 1/7 = 0.142857...
P5_genuine = 7 x q  where q ~ 10^11 * (1/7)
```

The 'boundary prime' is divisible by the very denominator that defines the KP convergence threshold. Whether this is a coincidence, a consequence of the construction of P5, or a deeper structural link to the C13 law -- this is an open question worth exploring (see Conjecture C7).

Required corrections (AUDIT-1 now closed)

Document	Original claim	Corrected status
TOWERS_YM_v2.3	P5_genuine = 1000000001119 is a 13-digit boundary prime	P5_genuine = 7 x 142857143017 is COMPOSITE (REJECTED). P5_replacement =
H4_Boundary.lean	C13 := 13 (boundary digit count, with P5_genuine as witness)	C13 := 13 is correct; witness replaced by P5_replacement = 1000000001083.
H4_Derivation.lean	C13_Law_Open: digit_len p >= 13 => Sym p = 1	Conjecture statement stands; P5_replacement is the corrected prime witness.

P5_replacement = 1000000001083: PRIME (Miller-Rabin, 12 witnesses), 13 digits, Sym=1 (omega=1 by primality). CERTIFIED.

Structural note: digit_sum(P5_replacement) = 13 = C13 boundary index.

certify_p5_replacement.py stdout SHA: 1799a8169bc5f62ba36fe17dcb7a547bb6cb6f7cde01704c8d41ef717ed7e80c

AUDIT-2: beta_0 bracket requires the full 36-moment computation

FINDING: Our 7-moment partial sum (m_0..m_6) cannot reproduce the CERT_Arb bracket. The partial sum converges too slowly near beta_0.

The CERT_Arb computation (in the source workspace, TheoremaAureum143) uses N=36 terms with all exact moments m_0..m_36, computed by constant-term extraction over the SU(3) torus. The tail bound is 4.46e-32, giving sufficient precision to locate the 1/7 crossing to 12 decimal places.

Our 7-moment partial sum underestimates w1 near beta_0 because the missing terms m_7..m_36 each contribute positive amounts. The qualitative result (w1(0.86) >> 1/7, D4 clearly fails) is confirmed even with 7 terms. The precise bracket requires the full computation.

Computation	Moments used	Precision near beta_0	D4 status	Verdict
Our script (7 terms)	m_0..m_6 (exact)	Insufficient (underestimates w1)	w1(0.86) >> 1/7: CONFIRMED	Bracket: NEEDS FULL CERT_ARB
CERT_Arb (authoritative)	m_0..m_36 (exact rational)	tail <= 4.46e-32 (dps=80)	w1(0.86)=0.432367... > 1/7	CERTIFIED: beta_0 in [2.079416880123,

CERT_Arb SHA: b5a9f0a7666a91f283a7d4531ae99dff2097c2cedef10424b77833b7bbc840d3

D4 failure (w1(0.86) >> 1/7) confirmed by our independent 7-term check.

---

14. What IS and IS NOT Proven -- Clay Rigour Checklist

WHAT IS PROVEN (machine-checked, trio-clean)

kp_summable_of_truncatedActivity: IF TruncatedActivityBound(a) AND N(n)<=7^n, THEN the entropy-weighted polymer series is summable. [Wall256_Note.lean]

mass_gap_pos_of_spectral_gap: rho^d = exp(-Delta*d) algebra. [Wall256_Note.lean]

strong_coupling_decay_of_open_inputs: threads H1+H2+H3 into abstract lattice decay. SORRY=0. Axioms={propext,Classical.choice,Quot.sound}. [Wall256_Scaffold.lean]

PeterWeyl_Summable_SU3: sum_{(m,n)} dim(m,n)^2*exp(-t*C2(m,n)) converges for t>0.

Truncation-to-envelope bridge S_N(t) <= K(t). [Tasks #154, #155]

beta_0 in [2.079416880123, 2.079416880124]: CERTIFIED (CERT_Arb, out-of-tower).

D4 at beta=0.86: w1(0.86) >> 1/7 CERTIFIED. D4 FAILS.

All 20 Bessel I_n(x) values verified against Z_BESSEL_I_COMPLETE CSV: ALL PASS.

Monotonicity of w1(beta): strictly decreasing (Laplace transform argument).

Strict-inequality threshold: I = log(7) is the exact divergence boundary.

S_N(t=1) convergence: series stabilises at N=5 to 6 decimal places.

P5_replacement = 1000000001083: PRIME, 13 digits, Sym=1 -- CERTIFIED. (P5_genuine = 1000000001119 was COMPOSITE; see AUDIT-1 correction, Section 13.)

WHAT IS NOT PROVEN -- explicit Clay-scope list

NO Yang-Mills mass gap. No mu>0 unconditionally. Delta>0 exists only under H1+H2+H3.

NO spectral gap for any real Wilson transfer operator. corr/sep are abstract; no real SU(3) operator built.

NO closure of Clay Surface #1. YM tower Status: OPEN.

NO continuum limit statement. Scope: LATTICE SU(3) strong-coupling only.

NO convergence of any real cluster expansion. KP conditional on H2's rate.

NO discharge of H1 (hw1), H2 (hOS), H3 (h_bridge). None scheduled.

NO cross-tower bridge between YM and NS reductions.

NO RH / GRH claims in YM certs. M7/GRH/C05 tower: SEPARATE.

P5_genuine: NOT PRIME. (AUDIT-1: = 7 x 142857143017.)

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15. Bessel I Zero-Pattern: LLM Error Topology on the Z-Protocol

This section reports an empirical study of how a language model (LLM) reproduces modified Bessel function values I_n(x) under the Z-protocol (T=0: LLM-direct, no computational tools; T=1: tool-assisted). The zero-error cells -- where the LLM gets the value exactly right -- are not randomly distributed. They trace out a specific locus determined by the Sym parameter and by the ratio x/n.

Experimental setup

Parameter	Values tested	Trials per cell
Order n	0, 1, 2, 3, 4, 5	5 (T=0 LLM-direct)
Argument x	0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 5.0, 10.0	5
Sym	1 (n=0,1); 2 (n=2,3,4,5)	--
Method T=0	A_LLM_T0: LLM answers with no tools	--
Method T=1	B_tool_T1: mpmath / rule echo	100 (all ZERO)
Source	BesselI_Z_MEASURE_(2) CSV	--

Zero map (T=0 LLM-direct): ZERO = correct / ERR = wrong / prt = partial

Function	x=0.5	x=1.0	x=1.5	x=2.0	x=2.5	x=3.0	x=5.0	x=10.0
I_0 (Sym=1)	ZERO	ZERO	ZERO	ZERO	prt	ZERO	ZERO	ZERO
I_1 (Sym=1)	ERR	ERR	ZERO	ERR	ERR	ZERO	ERR	ERR
I_2 (Sym=2)	ERR	ZERO	ERR	ERR	ERR	ERR	ERR	ERR
I_3 (Sym=2)	ERR	ERR	ERR	ERR	ERR	ERR	ERR	ERR
I_4 (Sym=2)	ERR	ERR	ERR	ERR	ERR	ERR	ERR	ERR
I_5 (Sym=2)	ERR	ERR	ERR	ERR	ERR	ERR	ERR	ZERO

Aggregate zero-error rate by Sym

Sym	n values	Total cells	LLM zeros (correct)	Full errors	Partial
1	n=0, n=1	16	9 (56.2%)	6 (37.5%)	1 (6.2%)
2	n=2..n=5	32	2 (6.2%)	30 (93.8%)	0 (0.0%)

The 11 zero-error cells and their x/n ratios

Cell	Sym	I_n(x) value (15 dp)	x/n ratio	Ratio type
I_0(0.5)	1	1.063483370667600	infty (n=0)	n=0: ratio undefined
I_0(1.0)	1	1.266065877752008	infty	n=0: all x rational
I_0(1.5)	1	1.646723189621999	infty	n=0: all x rational
I_0(2.0)	1	2.279585302336067	infty	n=0: all x rational
I_0(3.0)	1	4.880792585865024	infty	n=0: all x rational
I_0(5.0)	1	27.239871823604449	infty	n=0: all x rational
I_0(10.0)	1	2815.716628466254	infty	n=0: all x rational
I_1(1.5)	1	0.981666428761618	1.5 = 3/2	half-integer
I_1(3.0)	1	3.953370217402609	3.0 = 3/1	integer
I_2(1.0)	2	0.135747669767038	0.5 = 1/2	half-integer
I_5(10.0)	2	777.188286403312	2.0 = 2/1	integer

Key findings

T=1 (tool-assisted): 100% zero-error across ALL 48 cells. Tool use eliminates Bessel I reproduction errors entirely.

T=0, Sym=1: 56.2% zero-error rate. The LLM has internal representation of I_0 and I_1 across most of the tested range.

T=0, Sym=2: 6.2% zero-error rate. The LLM is almost entirely blind to I_n for n >= 2, except at two cells.

The two Sym=2 zero-error cells (I_2(1.0), x/n=0.5) and (I_5(10.0), x/n=2.0) both fall at integer or half-integer x/n ratios.

All non-n=0 zero-error cells (I_1, I_2, I_5) have x/n in {1/2, 3/2, 2, 3} -- rational numbers with small denominator.

The single partial-error cell I_0(2.5): x=2.5 is not a round number. 3 errors out of 5 (60% error rate) even for the best-known order n=0.

Interpretation

The LLM's zero-error locus is NOT random. It follows two rules:

RULE 1 (Sym): Sym=1 inputs (n=0,1) yield far more zeros than Sym=2 inputs. The model's internal Bessel knowledge is concentrated at low order.

RULE 2 (Ratio): Among higher-order functions (n >= 1), zeros appear only where x/n is a rational number with denominator 1 or 2. The model fails at irrational or non-simple-rational x/n points.

COMBINED: The LLM error function on Bessel I space has a zero-set that concentrates on Sym=1 AND rational x/n. This is a knowledge-topology map, not a random noise floor. An LLM operating without tools is reliable on I_0 (the most-tabulated order) but structurally unreliable on I_n (n >= 2) except at the simplest rational-ratio arguments.

Connection to the Wall256 / YM project

The Bessel I barrier (Open Question OQ-BI) requires formal I_n(x) bounds in Lean / Mathlib. The Z-protocol finding reinforces why: an LLM-only proof attempt on Bessel I would have a 93.8% error rate for Sym=2 inputs. The barrier is real. Tool-assisted computation (T=1) gives 0% error. The correct route is machine verification, not LLM recall.

Source: Bessell_Z_MEASURE_(2)_1780557621561.csv. All T=1 verification against Z_BESSEL_I_COMPLETE CSV: ALL 20 values PASS (relative error < 1e-10).

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16. Cryptographic Binding and Sources

Artifact	SHA-256 / Source
CERT_Arb_beta0 (interval certificate)	b5a9f0a7666a91f283a7d4531ae99dff2097c2cedef10424b77833b7bbc840d3
D4_w1_NEGATIVE_Certificate	9a794ccf0c707812e6fa3db2095a350f2d5b61a011fcc77e453c548716ac8764
D1_to_D3_Plan.md	a40422fe7595ebc2c3c7d8d4b80a2c5645431ac1a52f3699c732562d4574b553
Big_Day_Status_2026-06-01.md	9a00f1106bf93da155d8366521579eeec17bcacd2f4aea62fbcd0159dedfc7d1
hits.txt Genesis seal (460 bricks)	eecbcd9a540aa7a2c90edd23827c73e4d1bb5af641d352f70a5de849b21f875f
Lean commit (Wall256 scaffold)	8eeab54
Lean commit (Beta0Certified wiring)	ac241de

Per-file SHA-256 of the 13 YM/OS modules (from PDF 263, 2026-06-03)

#	File	SHA-256
1	YM/ClusteringCore.lean	f3ceecd3892edd81a760aa8205e947aef57ee273c84ef5be33c3300f8aa9a1fb
2	YM/MassGapStandin.lean	ffa6f5820e6d6fdc6b0af4051582be2508a35e211117d87e24f26eb799e8ee5d
3	YM/SpectralGapCore.lean	fca1de2b085e768a61ef320db8d7d23f546a0f3ddce0b76ccbc3f956eee69e89
4	YM/TransferOperatorBound.lean	ff0bb4887885a10e11ef0e41a46dd801acb0ff5cb58b75d3d613a3d27e10295c
5	YM/TwoPointDecay.lean	516b8fb721a762181a0200f11e8fd01f16ee01c19ae8228fca944ab907234575

6	YM/MassGapFromDecay.lean	b5acbbc0770f6211765989440ec82a88d6d2744ae7a5d45d383eaa1360adaa3b
7	YM/IntegratedTailReal.lean	223a0dacab5f7bc4b871b9a1dbe620d7944af091d7bbd65d6f6f693d68ef8ca2
8	YM/TransferGapReal.lean	bd30cd0f17d3a74ae8870c1fcaf2808a9584d6e77185274760367abda5484ebe
9	YM/MassGapReal.lean	30f0b1b37c510cee3265a1f5e3e2bcccbf04647a500273f28a09902ffcf51fb0
10	YM/ClusteringImpliesGap.lean	acc573093f6dc6b68e7496ebfd99932ba91a63937a5430c5f4e5eadea3ba1626
11	YM/TransferImpliesClustering.lean	3e502bacd4110ed8e52b5ad1841e98c0e45164e974756b013957b092bff7a301
12	YM/TailImpliesTransfer.lean	e0d3ba5d3cbcad9f17cea42a73cc155af00af2dbb91fc0254f70c8016643bd50
13	YM/GapToDecay.lean	ff4b8f5dd357a353982afe08e66afd8036d4eabc526af4d0c8c364828f235dbe